Centralizers, facets and finite amalgams on the projective line

Abstract

We give a common reduction for string C-groups of projective linear degree two. A nonabelian dihedral subgroup defined over an odd-characteristic subfield acquires no new centralizing projectivity under field extension. Combined with an odd-order normalizer argument, this excludes subfield facets without counting their mutual intersections. The remaining local markings form six classes; their fourteen compatible ordered pairs give ten universal presentations up to reversal. We retain Dickson's subgroup types and the group–polytope correspondence. The finite part is supplied by one exact computation. Separate applications derive the complete projective special and general linear principal classifications.

1. The common theorem

Let q be a prime power. We write $L(q) = \mathrm{PSL}_2(q)$ and $P(q) = \mathrm{PGL}_2(q)$. A string C-group is a group $G = \langle r_0, \dots, r_{n-1} \rangle$ with distinct involutory generators, with $r_i r_j = r_j r_i$ when $|i - j| > 1$, and with the intersection property

$$G_I \cap G_J = G_{I \cap J}, \qquad G_I = \langle r_i : i \in I \rangle. \tag{1}$$

The empty subgroup is trivial. In particular, no generator belongs to the subgroup generated by the others. The type of a string C-group is the sequence of orders of consecutive generator products.

Theorem 1 (uniform reduction). Suppose that a string C-group is isomorphic to $L(q)$ or $P(q)$ and has rank at least four. Then q is odd, the rank is four, and both end parabolics are proper and isomorphic to S_4 or A_5. Their ordered markings are among the six classes of Lemma 10. Up to reversal, the full marked group is a quotient of one of the ten literal presentations of Lemma 11.

The assertion is a reduction, not yet a classification of all quotients of those presentations. That last step belongs to the two application papers. The common theorem includes small fields and arbitrary subfield extension degrees. Its proof is completed in §6 after the general centralizer argument and the finite table have been established.

The six local classes and the ten-row table are earlier results from L083. The common proof below reworks their use in the two classifications. The centralizer calculation and the normalizer assembly replace the source's subfield-incidence counts. No claim of historical priority for this proof method is needed for the applications.

2. Retained inputs and elementary string consequences

We use two established background theorems.

Input D (Dickson's classification, in its projective form). For a finite field K of odd characteristic, a proper subgroup of $L(K)$ or $P(K)$ is one of the following types: cyclic; dihedral; elementary abelian of characteristic order; a subgroup of an affine group $V \rtimes k^\times$; A_4, S_4, or A_5; or a conjugate of $L(k)$ or $P(k)$ for a subfield $k \subseteq K$. Only the types that actually embed are intended. The subfield alternatives are conjugate to the usual matrix embeddings, not merely abstractly isomorphic groups. We also retain the usual orders

$$|L(q)| = \frac{q(q^2-1)}{(2, q-1)}, \qquad |P(q)| = q(q^2-1), \tag{2}$$

the simplicity of $L(q)$ for $q \geq 4$, and the trivial centralizer of $L(q)$ in $P(q)$.

The required subgroup list is printed as Theorem 2 in L083 and as Propositions 1–2 in L079, with attribution there to Dickson, Moore and the later projective subgroup account. Their subgroup multiplicity formulas are not inputs to our proof. In particular the affine alternative includes dihedral groups of order $2p$, where $p = \mathrm{char}\, K$; it must not be silently replaced by the semisimple torus cases.

Input P (the string C-group correspondence). String C-markings, up to group isomorphisms preserving the ordered marking, correspond to regular abstract polytopes. The faces of rank i are the cosets of the subgroup generated by all generators except r_i. Reversing the marking gives the dual polytope. This is the standard correspondence used in both sources and in the existing proof library. A group epimorphism preserving the ordered distinguished generators between two string C-groups induces a covering of their regular polytopes. We retain this covering part of the correspondence as well.

We first dispose of a possible degeneracy. If a consecutive product has order two, all generators on one side of that place commute with all generators on the other. By (1) the two generated subgroups meet trivially. Since they generate G, the multiplication map from their direct product to G is an isomorphism. Thus a directly indecomposable group has only types with consecutive entries at least three.

All the groups $L(q)$ and $P(q)$ are directly indecomposable. For $q \geq 4$, simplicity proves this for $L(q)$. If $P(q) = A \times B$, its simple normal subgroup $L(q)$ must lie in one factor: its projections have commuting normal images, and a simple nonabelian group cannot have two nontrivial such images. More explicitly, $L(q) \cap A$ and $L(q) \cap B$ are normal, and if both were trivial, the commutators of $L(q)$ with both factors would be trivial. The other factor then centralizes $L(q)$, so is trivial. In even characteristic $P(q) = L(q)$. The remaining groups are S_3, A_4, S_4, for which the elementary normal-subgroup lists give the same conclusion. These lists are $1, A_3, S_3$; $1, V_4, A_4$; and $1, V_4, A_4, S_4$, respectively.

Every marking of rank at least four in these groups therefore has all consecutive product orders at least three. Restricting a marking to an interval of three indices gives an irreducible rank-three string C-group.

Lemma 2 (normalizing a facet). Suppose $G = \langle H, s \rangle$, $s^2 = 1$, and $sHs^{-1} = H$. If $t \in H$ and ts has odd order, then $s \in H$.

Proof. The normalizer assumption makes $H \lhd G$. The quotient is generated by sH, whose order divides two. Since $tH = H$, the order of the same element $tsH = sH$ divides the odd order of ts. Hence $sH = H$. □

This lemma is valid without a finiteness assumption on G, provided the displayed odd order exists. It is the common final step in both subfield exclusions.

3. Centralizers on the projective line

We prove the matrix facts used in the reduction. All projectivities act on the projective line; equality of matrices representing them is understood up to nonzero scalar.

Lemma 3 (characteristic two). In $P(K)$, for K of characteristic two, the centralizer of every nonidentity involution is abelian.

Proof. Over an algebraic closure an involution is unipotent, since a semisimple element of order two would have eigenvalue ratio $-1 = 1$. Conjugate it to $u : x \mapsto x + 1$. It has the unique fixed point infinity, so a commuting projectivity must fix infinity and has the form $x \mapsto ax + b$. Commutation gives $a = 1$. These translations commute with one another. The centralizer over K is a subgroup of this abelian group. □

Lemma 4 (commuting involutions in odd characteristic). Let K have odd characteristic. The centralizer of a four-group in $P(K)$ is that four-group. In particular $P(K)$ contains no elementary abelian subgroup of order eight.

Proof. We may extend scalars to an algebraic closure. Put the first involution in the form $z : x \mapsto -x$. A projectivity commuting with z either preserves or interchanges its two fixed points, and accordingly has the form $x \mapsto ax$ or $x \mapsto a/x$. The only nontrivial involution of the first form is z. An independent commuting involution is therefore of the second form; rescaling the coordinate makes it $w : x \mapsto 1/x$.

The projectivities commuting with both z and w are exactly

$$x, \quad -x, \quad 1/x, \quad -1/x.$$

These are the given four-group. Since its elements were already defined over K, intersecting with $P(K)$ changes nothing. □

Here is the field-extension statement responsible for the common reduction.

Theorem 5 (dihedral centralizer descent). Let $k \subseteq K$ be fields of odd characteristic. Let $a, b \in P(k)$ be distinct involutions such that $u = ab$ has finite order greater than two. Then

$$C_{P(K)}(\langle a, b\rangle) \subseteq P(k). \tag{3}$$

If u is unipotent, this centralizer is trivial. Otherwise it has order two. In the semisimple case, if $A \in \mathrm{GL}_2(k)$ represents u, its nontrivial element is represented by

$$Z = A - \tfrac{1}{2}\,\mathrm{tr}(A)I. \tag{4}$$

Proof. First suppose u is semisimple. Over an algebraic closure it has two distinct eigenlines, and we may write it as $x \mapsto \lambda x$, where $\lambda \neq \pm 1$. A projectivity commuting with u must preserve its individual fixed points: an interchange would conjugate u to $u^{-1} \neq u$. Its form is therefore $x \mapsto cx$. Since $aua^{-1} = u^{-1}$, the involution a interchanges the eigenlines and has the form $x \mapsto d/x$. Commutation of $x \mapsto cx$ with a gives $c^2 = 1$. Thus the centralizer over the algebraic closure is precisely $\{1, z\}$, where $z : x \mapsto -x$.

Write the eigenvalues of A as α, β. The matrix in (4) has eigenvalues $(\alpha - \beta)/2$ and $(\beta - \alpha)/2$. Both are nonzero, and their ratio is -1. Thus (4) represents z, is invertible, and has entries in k. This proves both the exact order and (3), including the nonsplit torus case. The expression is unaffected projectively if A is rescaled.

Now suppose u is unipotent. Over the algebraic closure put $u : x \mapsto x+1$. As in Lemma 3 its centralizer consists of translations $x \mapsto x + c$. An involution conjugating u to u^{-1} has form $x \mapsto -x + d$. A translation commuting with it satisfies $2c = 0$, hence $c = 0$. The centralizer is trivial. These two cases exhaust nonidentity elements of $P(k)$ in positive characteristic. □

The unipotent case explains directly why a dihedral group of order $2p$ cannot have an independent commuting involution in odd characteristic. This is the case singled out in the corrective remark following Proposition 2 of L079. No absence of dihedral groups of order $2p$ is assumed here.

Corollary 6 (odd end labels). In a rank-four string C-group contained in $P(K)$, with K of odd characteristic and all consecutive labels at least three, the orders of r_0r_1 and r_2r_3 are odd.

Proof. Put $D = \langle r_0, r_1\rangle$ and let $t = |r_0r_1|$. The generator r_3 centralizes D and lies outside it by (1). Theorem 5 first excludes the unipotent case. If t were even, the nontrivial involution $(r_0r_1)^{t/2} \in D$ would centralize D. The centralizer has exactly one nonidentity element, so it would equal r_3, a contradiction. Reverse the indices for the other end label. □

4. The uniform rank and facet reduction

Proposition 7 (rank restriction). A string C-group isomorphic to $L(q)$ or $P(q)$ has rank at most four. If its rank is four, q is odd.

Proof. In odd characteristic a marking of rank at least five contains the independent commuting involutions r_0, r_2, r_4. Their generated subgroup is elementary abelian of order eight: (1) excludes every dependence. This contradicts Lemma 4.

In characteristic two, a marking of rank at least four has $r_2, r_3 \in C_G(r_0)$. Lemma 3 implies that they commute, contrary to the consecutive label being at least three. This also excludes all higher ranks. □

Lemma 8 (the nonexceptional rank-three groups). An irreducible rank-three string C-group cannot be cyclic, elementary abelian, dihedral, A_4, or a subgroup of the one-dimensional affine group in odd characteristic.

Proof. Cyclic and elementary abelian groups are immediate. The involutions of A_4 generate its four-group and cannot generate A_4.

In an affine group in odd characteristic, an involution must have multiplier -1, so has form $x \mapsto -x + v$. Two such involutions commute only when their translation parameters agree. Thus there cannot be two distinct commuting end generators.

For a dihedral group use its usual rotation/reflection description. A central end generator would give an adjacent label two. Otherwise both end generators are reflections. Two distinct commuting reflections have product the central involution z, so $r_2 = zr_0$. The middle generator cannot be central, again by irreducibility, and hence is a reflection. Put $u = r_0 r_1$. Then

$$(r_1 r_2)^2 = (u^{-1} z)^2 = u^{-2}.$$

The nonidentity rotation u^2, nonidentity because $|u| > 2$, belongs to both adjacent rank-two subgroups. It does not belong to the reflection subgroup $\langle r_1 \rangle$. This violates (1). □

Theorem 9 (exceptional facets). Let $G \cong L(q)$ or $P(q)$ have a rank-four string C-marking. Its facet and vertex-figure groups are isomorphic to S_4 or A_5, and are proper subgroups.

Proof. Proposition 7 gives odd characteristic. Let $H = \langle r_0, r_1, r_2 \rangle$. It is proper by (1) and irreducible by §2. Input D and Lemma 8 leave only S_4, A_5, and the conjugates of standard subfield groups $L(k)$ and $P(k)$.

Suppose H is a subfield group, and conjugate the embedding so that it is standard. The two involutions $r_0, r_1 \in H \subseteq P(k)$ have product order greater than two. Theorem 5 places their centralizer in $P(k)$. Since r_3 centralizes them, $r_3 \in P(k)$. If $H = P(k)$, this immediately contradicts (1). If $H = L(k)$, the determinant-square homomorphism shows that $P(k)$ normalizes H. Hence r_3 normalizes H. Corollary 6 makes $|r_2 r_3|$ odd, so Lemma 2, with $t = r_2$, again gives $r_3 \in H$, a contradiction. Reversing the marking proves the vertex-figure assertion. □

No comparison of the extension degrees is needed. This proof includes subfield embeddings of degree two, arbitrarily large extension degrees, and the small subfield groups whenever the subgroup list permits them. Exceptional groups abstractly isomorphic to $P(3)$ or $L(5)$ in a *different* characteristic are not treated as subfield embeddings; they correctly remain among S_4, A_5.

5. The exceptional finite calculation

For integers m, n, k, write $[m, n]_k$ for the presentation

$$\langle x, y, z \mid x^2 = y^2 = z^2 = (xz)^2 = (xy)^m = (yz)^n = (xyz)^k = 1 \rangle. \tag{5}$$

The corresponding map, when it is a string C-group, is traditionally written $\{m, n\}_k$. This is the Petrie relation convention of L083, physical page 9.

Finite Lemma 10. Up to automorphisms preserving the ordered marking, the irreducible rank-three string C-markings of S_4 and A_5 have precisely the following triples (m, n, k):

$$\begin{array}{c|ccc} S_4 & (3,3,4) & (3,4,3) & (4,3,3) \\ A_5 & (3,5,5) & (5,3,5) & (5,5,3). \end{array} \tag{6}$$

In each case (5) is a presentation of the indicated group.

Finite proof and audit procedure. The program enumerates every involution of S_4 and A_5, every ordered pair of distinct commuting involutions for x, z, and every middle involution y. It tests generation, both adjacent orders, and (1) on all eight subsets. There are 72 surviving ordered triples in S_4 and 360 in A_5. It partitions these by the full automorphism group, not merely inner conjugation, and obtains the three ordered types in each row of (6). It separately constructs (5) as a finitely presented group and checks order 24 or 60. A marking gives a surjection from (5) onto the corresponding finite group; equality of orders makes it an isomorphism. These are exhaustive finite calculations with GAP's exact subgroup and finite-presentation machinery. □

For two entries $f = (m, n, k)$ and $v = (n, s, l)$ from (6), define

$$\begin{aligned} U(f,v) = \langle r_0, r_1, r_2, r_3 \mid & r_i^2 = 1,\ (r_i r_j)^2 = 1\ (j - i > 1), \\ & (r_0 r_1)^m = (r_1 r_2)^n = (r_2 r_3)^s = 1, \\ & (r_0 r_1 r_2)^k = (r_1 r_2 r_3)^l = 1 \rangle. \end{aligned} \tag{7}$$

The marking of every rank-four polytope with these local maps is a quotient of (7). Matching the common middle label in (6) gives fourteen ordered pairs. Reversal sends (f, v) to (v^*, f^*), where $(a, b, c)^* = (b, a, c)$. There are ten orbits, including six fixed pairs; the following table contains one from each orbit.

Row	Facet f	Vertex figure v	$\lvert U(f,v)\rvert$	Group	String C-marking
1	$(5,5,3)$	$(5,5,3)$	1	trivial	no
2	$(5,5,3)$	$(5,3,5)$	1	trivial	no
3	$(5,3,5)$	$(3,5,5)$	3420	$L(19)$	yes
4	$(5,3,5)$	$(3,4,3)$	60	A_5	no
5	$(5,3,5)$	$(3,3,4)$	1	trivial	no
6	$(4,3,3)$	$(3,4,3)$	96	$2^4 : S_3$	yes
7	$(4,3,3)$	$(3,3,4)$	24	S_4	no
8	$(3,5,5)$	$(5,3,5)$	660	$L(11)$	yes
9	$(3,4,3)$	$(4,3,3)$	1	trivial	no
10	$(3,3,4)$	$(3,3,4)$	120	S_5	yes

Finite Lemma 11. The table is correct. The groups in rows 3, 8 and 10 admit automorphisms reversing their four specified generators. The face counts in these rows, from vertices through facets, are respectively

$$(57, 171, 171, 57), \quad (11, 55, 55, 11), \quad (5, 10, 10, 5). \tag{8}$$

The graph of vertices and edges in row 8 is the complete graph K_{11}.

Finite proof and audit procedure. For each literal presentation (7), the program computes the order of the finitely presented group, constructs an isomorphic permutation group, and transports each specified generator through that isomorphism. It tests (1) on all sixteen subsets, including nonconsecutive ones. Thus a group identification alone does not certify a polytope. The identity group is handled separately, since GAP returns `fail` for its generic finitely presented permutation-isomorphism request.

The three main rows are identified by an actual group isomorphism with $L(19), L(11), S_5$, respectively. Reversal is checked by constructing the homomorphism on the four generators and checking bijectivity. The program computes the four maximal-parabolic indices, giving (8). For row 8 it constructs the actual vertex and edge cosets and their incidences. Every edge has two endpoints and the resulting 55 unordered pairs are all distinct, proving the K_{11} assertion.

For the auxiliary row 6 it verifies an elementary abelian normal subgroup of order 16, quotient S_3, and the existence of a complement. The indicated marking has face counts $(4, 6, 6, 4)$. The failures in rows 4 and 7 are checked by the intersection test; independently, a stipulated facet or vertex-figure group already has the order of the full group, so it cannot occur as the required proper parabolic. The remaining failures are trivial groups. □

This calculation recovers the actual ten-row table of L083, rather than selecting only its two successful linear-group rows. The six local maps and ten amalgams are prior results; the program supplies a new replay of them.

6. Proof of the reduction and verification

Proof of Theorem 1. Section 2 excludes consecutive labels two, including in the small groups. Proposition 7 then gives rank four in odd characteristic. Theorem 9 identifies both proper end parabolics as S_4 or A_5. Lemma 10 exhausts their ordered markings and supplies an exact presentation for each local group. The consecutive pairs share their middle product order. Substituting their two presentations into the four-generator string relations gives the marked epimorphism from (7). All compatible pairs, with their reversals, occur in Lemma 11. This proves every assertion. □

The common mathematical proof consists of this paper and the single program `phase230_finite_check.g`. The finite-presentation computations use GAP's exact algorithms to obtain the order of each *presented group*, and then an isomorphism with a permutation group. Checking a chosen permutation image alone would give only a lower bound and would not justify the table. Here the exact computation also verifies all marked subgroup intersections, group identifications, self-dualities and the listed coset incidences. The same program is used by both applications; it is not rerun as an additional mathematical hypothesis for either.

The inherited formal file `phase230_formal/DihedralFacetObstruction.lean` proves five abstract steps: an involution with an odd vanishing power is trivial; the corresponding homomorphism assertion; membership in a normal facet; derivation of this conclusion from the normalizer condition and generation; and contradiction with an independent generator. These are parts of Lemma 2 and its use. They do not formalize the matrix centralizers, Dickson's theorem, the finite presentations, or Theorem 1 as a whole. The finite program and those remaining written proofs retain their stated verification status.

A portable replay verifies the six local classes and all ten universals. Changing a Petrie relator is rejected by the universal-order check; duplicating a distinguished generator is rejected by the full intersection test. The formal replay reports only the usual Lean axioms. The accompanying contract gives the exact files, versions and retained source editions.

7. Attribution and the application interface

Dickson's subgroup classification, the ordinary projective group structure facts and the string C-group/regular-polytope correspondence are retained inputs, not new results. The six exceptional rank-three maps and the amalgam table are recovered from L083 with an explicit replay. The classifications derived in the application papers belong to Leemans and Schulte; the 11-cell and 57-cell themselves are the earlier constructions of Grünbaum and Coxeter.

Relative to the existing proof bank, the work here is the explicit projective-line centralizer descent and its common assembly with the odd-order normalizer argument. An analogous centralizer strategy already appears in the retained Suzuki discussion; it is not a substitute for the matrix proof in §3. The present paper preserves the complete common proof from the independently reviewed combined manuscript, while the two applications carry their separate quotient and geometric obligations.

The applications use Theorem 1 and the exact marking data of Lemma 11. Neither application is a premise of the other. Together they recover two principal classification contracts; they do not reprove

every secondary statement in the historical sources. In particular the almost-simple seventeen-case census and the separate conjecture in L079 are not consequences asserted here. The topic's cost includes this entire common paper and its required finite program once, followed by each complete application. A common lemma used in another topic is charged again as part of that other topic's full chosen umbrella.

References

[L083] D. Leemans and E. Schulte, *Groups of type $L_2(q)$ acting on polytopes*, retained arXiv:math/0606660v1 author edition, 14 physical pages; *Advances in Geometry* 7 (2007), 529–539. Theorem 2 prints the retained subgroup classification; §5 prints the exceptional amalgam table.

[L079] D. Leemans and E. Schulte, *Polytopes with groups of type $PGL_2(q)$*, retained arXiv:0909.1991v1, 9 physical pages. Propositions 1–2 print the projective subgroup inputs; Theorem 10 is the principal projective general linear classification.

[MS] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002. The marked-group correspondence, face cosets, duality and marked epimorphism/covering theorem are retained.